global and random token attentions in the encoder.

For encoder variants, we pick the best performing model from fixed patterns to be combined with full encoder-decoder attention, i.e., sliding window with stride (STRIDE), low-rank method (LIN.), and learnable patterns (LSH and SINKHORM). We then combine learnable patterns with HEPOS to support processing more text. All models consume as long an input as the memory allows.

Results. Overall, *models that read more text obtain higher ROUGE scores*, according to results on GovReport and PubMed in Table 4. First, different encoder variants with full encoder-decoder attentions attain better results than the full attentions baseline except Linformer. Second, *adding* HEPOS *encoder-decoder attention almost doubles the words that can be processed and further improves the performance.* This highlights the importance of handling both encoder attentions and encoder-decoder attentions efficiently. Notably, HEPOS with an LSH encoder *achieves new state-of-the-art results on PubMed*, outperforming BigBird which only uses sparse attentions on the encoder. We also report performances of our two best models with HEPOS on arXiv in Table 5, and they outperform all competitive abstractive models.

As can be seen from the sample summaries in Fig. 3, our model that reads in 10k tokens generates more informative summary than the full attention model that only processes 1k tokens. Fig. 4 further shows that ROUGE-2 scores can be consistently lifted when reading more input, with similar trends observed on ROUGE-1 and ROUGE-L. More sample outputs are presented in Appendix C.

6.4 Reading More Input Improves Faithfulness

Here we first show **human evaluation** results on informativeness and unfaithful errors in the generated summaries. We sample 100 documents from GovReport and PubMed (50 each) with structured references that are labeled with aspects as described in § 4 and Appendix D. Each sample is evaluated by two fluent English speakers, who have cumulatively annotated tens of thousands of sentences for the same tasks before this work. Annotators are asked to label each summary sentence with an aspect and then decide whether it contains any type of error. Three types of unfaithful errors are considered: (i) **hallucination**—fabricating content not present in the input, (ii) **deletion**—incorrectly deleting crucial entities, events, or clauses, and (iii) **false concatenation**—inappropriately concatenating components from different sentences. 1 is given if any judge determines that a certain type of error exists in the sentence, 0 otherwise.

After reading the full summaries, each judge also scores aspect-level **informativeness**—whether the

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Human-written Summary:
In fiscal year 2018, Medicaid covered approximately 75 million individuals at an estimated cost of $629 billion, $393 billion of which were federal funds. (...)
While CMS is generally required to disallow, or recoup, federal funds from states for eligibility-related improper payments if the state's eligibility error rate exceeds 3 percent, it has not done so for decades, because **the method it used for calculating eligibility error rates was found to be insufficient for that purpose.** To address this, in July 2017, CMS **issued revised procedures through which it can recoup funds for eligibility errors**, beginning in fiscal year 2022. (...)

Model w/ full attn.:
Medicaid is a federal-state program that provides health care coverage to low-income individuals and families. (...)
CMS officials stated that **they have provided states with guidance** on how to use data from SSA's automated system for eligibility determinations, (...)
CMS officials said that **they did not have guidance** on when states should use SSA data to evaluate eligibility based on nonfinancial or financial criteria. (...)

Model w/ HEPOS enc-dec attn. (ours):
The Patient Protection and Affordable Care Act (PPACA) expanded Medicaid coverage to millions of low-income adults and children with disabilities and their eligible dependents. (...)
The selected states also reported that **they did not have adequate processes to address these issues.** CMS has taken steps to **improve its oversight of the Medicaid program, including issuing guidance to states** on the use of MAGI-exempt bases for determining eligibility, but these efforts have not been fully implemented. (...)

Figure 3: Sample summaries for a government report. The model with truncated input generates **unfaithful content**. HEPOS attention with a Sinkhorn encoder covers **more salient information**.

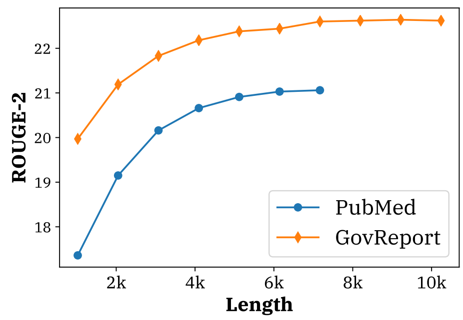

Figure 4: Summarizing articles truncated at different lengths by the best models: LSH (7168)+HEPOS on PubMed and SINKHORN (10240)+HEPOS on GovReport. Reading more consistently improves ROUGE-2.

System (MaxLen)	Inf.↑	Hal.↓	Del.↓	Concat.↓
GovReport				
Encoder variants w/ full enc-dec attn.				
FULL (1024)	3.29	15.2%	3.5%	9.5%
SINKHORN (5120)	3.32	**11.0%**	**2.3%**	9.4%
Encoder variant w/ HEPOS enc-dec attn. (ours)				
SINKHORN (10240)	**3.53**	11.5%	3.4%	**8.8%**
PubMed				
Encoder variants w/ full enc-dec attn.				
FULL (1024)	3.27	20.1%	2.8%	14.3%
SINKHORN (5120)	3.94	4.8%	**1.6%**	9.6%
Encoder variant w/ HEPOS enc-dec attn. (ours)				
SINKHORN (10240)	**4.18**	**3.5%**	2.2%	**9.1%**

Table 6: Human evaluation on informativeness (Inf.) (1-to-5), and percentages of unfaithful errors due to hallucination (Hal.), deletion (Del.), and false concatenation (Concat.). Inter-rater agreement with Krippendorf's α for all columns: 0.59, 0.59, 0.53 and 0.60.

summary covers important information of an aspect when compared with the reference. All system summaries and references are presented in a random order. Human evaluation guidelines and sample summaries for different aspects are included in Appendix D.

Results. Overall, *reading more text significantly improves informativeness as well as reduces fabricated content.* From Table 6, we observe that HEPOS attention, combined with a SINKHORN encoder, obtains better informativeness scores than comparisons that read in less text on both datasets. This echos results from automatic evaluation in the previous section. Moreover, both models that use efficient attentions reduce unfaithfulness, especially hallucination errors, when compared with the full attention model, which only reads 1024 tokens. As the models read more content, they learn to surface more factual and richer content in the summaries, as seen in Fig. 3.

Next, we explore if reading more helps correctly reflect the content in documents' later sections. We plot aspect-level human ratings of informativeness and unfaithful errors on PubMed and GovReport in Fig. 5 and Fig. 6. We report percentages of sentences with unfaithful errors by majority voting (i.e., at least one error is found by both annotators in the sentence). As can be seen, our models consistently improve informativeness and reduce errors across sections, especially for "Results" and "Conclusions" on PubMed and "What GAO recommends" on GovReport—these sections often appear in the later part of the source documents.

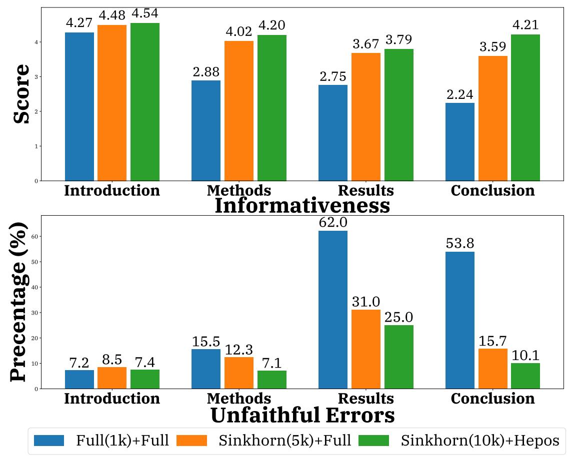

Figure 5: Aspect-level informativeness and percentages of sentences containing unfaithful errors as labeled by both human judges on PubMed. Models with efficient attentions reduce errors for later sections in the sources, e.g., "Results" and "Conclusion".

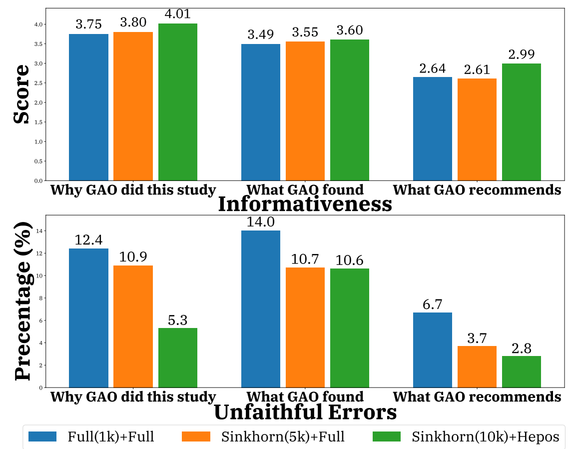

Figure 6: Aspect-level informativeness and percentages of sentences with unfaithful errors on GovReport.

Especially, we find that the full attention model tends to produce fabricated numbers in resultant summaries, whereas our models are able to correct them.

Lastly, we report the entailment-based FactCC and QA scores APES and APES$_{src}$ for top performing models in Table 7. The results again show that *consuming longer input leads to more faithful summaries*, though the differences are less pronounced.

6.5 Correlations between Human and Automatic Metrics

Finally, we study whether the faithfulness evaluation metrics correlate with human judgment. As shown in Table 8, on both government reports and scientific papers, *QA metrics are better correlated with human ratings, with our newly pro-*

System (MaxLen)	GovReport F.	APES	APES$_{src}$	PubMed F.	APES	APES$_{src}$
FULL (1024)	58.9	42.7	42.7	**74.6**	43.2	31.5
Encoder variants w/ full enc-dec attn.						
STRIDE (4096)	55.3	43.1	42.5	72.7	43.8	31.9
LIN. (3072)	48.4	35.7	36.3	67.7	39.3	29.5
LSH (4096)	55.7	**44.0**	43.6	73.2	46.7	35.1
SINKHORN (5120)	57.0	43.6	42.1	72.9	46.8	35.4
Encoder variants w/ HEPOS enc-dec attn. (ours)						
LSH (7168)	59.6	**44.0**	44.2	73.3	**47.5**	**35.6**
SINKHORN (10240)	**60.1**	**44.0**	**44.3**	71.9	46.2	34.8

Table 7: Evaluation with FactCC (F.), APES, and the new APES$_{src}$ metric, with higher numbers indicating more faithful summaries.

Metric	GovReport Inf.↑	Err.↓	PubMed Inf.↑	Err.↓
FactCC	0.07	-0.08	0.10	-0.14
APES	0.16	-0.15	0.25	-0.31
APES$_{src}$	**0.21**	**-0.23**$*$	**0.32**$*$	**-0.32**

Table 8: Pearson correlation between human ratings and metrics. We use aggregated unfaithful errors (Err.). $*$: significantly better than other metrics based on William's test (Williams, 1959) ($p < 0.05$).

posed APES$_{src}$ *being the stronger of the two.* After inspection, we find that human-written summaries contain paraphrases or acronyms that APES cannot capture via strict lexical matching. For instance, for the question "Diabetes may worsen ___ in patients", the reference answer is "death rate", whereas answers from the source and the system summary are both "mortality". APES$_{src}$ captures this, but not APES.

7 Additional Related Work

Summarizing long inputs has been investigated in many domains, including books (Mihalcea and Ceylan, 2007), patents (Trappey et al., 2009), movie scripts (Gorinski and Lapata, 2015), and scientific publications (Qazvinian and Radev, 2008). However, the datasets are often too small to train neural models. Cohan et al. (2018) publish two large-scale datasets by collecting articles from ARXIV and PUBMED. Popular methods rely on extractive summarizers that identify salient sentences based on positional information (Dong et al., 2020) or combined global and local contexts (Xiao and Carenini, 2019), where each sentence is represented as aggregated word embeddings. However, extractive summaries are often redundant and incoherent, highlighting the need for handling long documents via abstractive summarization.

To that end, extract-then-abstract methods are proposed. For example, Pilault et al. (2020) first extract relevant sentences and then rewrite them into paper abstracts. Our work is in line with building end-to-end abstractive summarization models for long input. Cohan et al. (2018) design a hierarchical encoder to read different sections separately, and then use combined attentions over words and sections to generate the summary. Multiple agents are created to read segments separately, and then collaboratively write an abstract (Celikyilmaz et al., 2018). However, both work truncates articles to 2K words. Although efficient encoder attentions have been studied in Zaheer et al. (2020) for abstractive summarization, at most 3K tokens can be consumed by their models. Our HEPOS encoder-decoder attention are able to process more than 10K tokens, significantly improving summary informativeness and faithfulness.

8 Conclusion

We investigate efficient attentions for long document summarization. We propose a novel encoder-decoder attention, HEPOS, based on head-wise positional strides that can effectively identify salient content. Models based on HEPOS attention can process at least twice as many words and produce more informative summaries with less unfaithful errors, according to both automatic evaluation and human evaluation. We further show that our new cloze QA metric better correlates with human judgment than prior faithfulness evaluation metrics.

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